

Fashion MNIST CNN Classifier - Jupyter Notebook Code

1. Install Required Libraries

```
!pip install tensorflow matplotlib numpy
```

2. Import Libraries

```
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow import keras
import numpy as np
```

3. Load Fashion MNIST Dataset

```
(x_train, y_train), (x_test, y_test) = keras.datasets.fashion_mnist.load_data()
```

4. Display Images

```
plt.imshow(x_train[1])
plt.show()

plt.imshow(x_train[0])
plt.show()
```

5. Normalize Dataset

```
x_train = x_train.astype('float32') / 255.0
x_test = x_test.astype('float32') / 255.0
```

6. Reshape Dataset

```
x_train = x_train.reshape(-1, 28, 28, 1)
x_test = x_test.reshape(-1, 28, 28, 1)
```

7. Check Shapes

```
print(x_train.shape)
print(x_test.shape)

print(y_train.shape)
print(y_test.shape)
```

8. Build CNN Model

```
model = keras.Sequential([

    keras.Input(shape=(28,28,1)),

    keras.layers.Conv2D(
        32,
        (3,3),
```

```

        activation='relu'
    ),

    keras.layers.MaxPooling2D((2,2)),
    keras.layers.Dropout(0.25),

    keras.layers.Conv2D(
        64,
        (3,3),
        activation='relu'
    ),

    keras.layers.MaxPooling2D((2,2)),
    keras.layers.Dropout(0.25),

    keras.layers.Conv2D(
        128,
        (3,3),
        activation='relu'
    ),

    keras.layers.Flatten(),

    keras.layers.Dense(
        128,
        activation='relu'
    ),

    keras.layers.Dropout(0.25),

    keras.layers.Dense(
        10,
        activation='softmax'
    )

])

```

9. Model Summary

```
model.summary()
```

10. Compile Model

```

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy'] )

```

11. Train Model

```
history = model.fit(  
    x_train,  
    y_train,  
    epochs=10,  
    validation_data=(x_test, y_test)  
)
```

12. Evaluate Model

```
test_loss, test_acc = model.evaluate(x_test, y_test)  
print('Test accuracy:', test_acc)
```